

CH 11 53

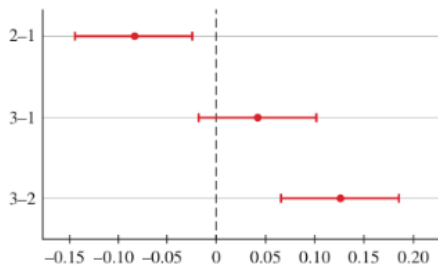
14. Calcium, again Refer to Exercise 13 (Section 11.2), where the calcium content of a powdered mineral substance was analyzed using three different methods. The comparisons option in MINITAB provided the printout that follows.

Tukey Simultaneous Test for Differences of Means

Difference of Levels	Difference of Means	SE of Difference	95% CI	T-Value	Adjusted P-Value
2-1	-0.0840	0.0224	(-0.1438, -0.0242)	-3.75	0.007
3-1	0.0420	0.0224	(-0.0178, 0.1018)	1.87	0.189
3-2	0.1260	0.0224	(0.0662, 0.1858)	5.62	0.000

Individual confidence level = 97.94%

**Tukey Simultaneous 95% CIs
Differences of Means for Calcium**



If an interval does not contain zero, the corresponding means are significantly different.

Use Tukey's method to compare the three population means. How would you summarize your results?

2-1. ① 95% CI doesn't include 0

⇒ we have sufficient evidence to say that 2-1 has significant difference

② p-value = 0.007 < 0.05 ⇒ 2-1 has significant difference

3-1. ① 95% CI include 0

⇒ we have sufficient evidence to say that 3-1 has same mean

② p-value = 0.189 > 0.05 ⇒ 3-1 doesn't have significant difference

3-2. ① 95% CI doesn't include 0

⇒ we have sufficient evidence to say that there is difference between 3-2.

② p-value = 0.000 < 0.05 ⇒ 3-2 have significant difference